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# Inorganic salt induced thermal reversible and anti-freezing cellulose hydrogels

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Dedication ((optional))

**Abstract:** Inspired by the anti-freezing mechanisms found in nature, ionic compounds ($ZnCl_2$/$CaCl_2$) are integrated into the cellulose hydrogel networks to enhance the freezing resistance. In this work, the cotton cellulose is dissolved by a specially designed $ZnCl_2$/$CaCl_2$ system, which endows the cellulose hydrogels specific functionality such as excellent freeze-tolerance, good ion conductivity and superior thermal reversibility. Interestingly, the coagulation speed of cellulose could be promoted by the addition of extra water or glycerol. This new type of cellulose based hydrogels is expected to construct flexible devices for services at temperature as low as -70 °C.

The water-rich structures of hydrogels are similar to biological tissues.[1] The liquid-like transport behavior and solid-like mechanical properties bestow the hydrogels diverse applications.[2] Recently, the hydrogels have been emerged as desirable materials for wearable devices, flexible electrodes, tissue engineering, wound dressings and drug delivery owing to their good toughness, optical transparency, and high conductivity.[3] However, when the temperature falls below the water freezing point (0 °C), the majority of hydrogels composed of hydrophilic polymers inevitably freeze.[4] The hydrogels would become fragile and loss their pristine elasticity. Therefore, there is an urgent need to develop anti-freezing hydrogels which can work under a broad range of temperature.

In nature, a number of organisms can tolerate cold conditions. For example, rainbow smelt and peeper frog have evolved means to survive extreme cold by producing salts or alcohols in their body.[5] The exceptional freezing tolerance derives from the colligative property of high concentrated compounds, which suppress the generation of ice crystals in cells.[6] This freezing-point depression phenomenon is what causes seawater (a mixture of salts in water) to remain liquid at subzero temperatures.

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Road salting takes advantage of this effect to protect roads from icing over and salts are widely applied as inhibitor for water freezing. Commonly used NaCl can depress the freezing point of pure water to about -21 °C.[4b] Inspired by the anti-freezing mechanisms found in nature, Jiang et al. designed a hydrophilic/oleophilic heteronetwork to construct mechanically strong organohydrogels with outstanding freeze-tolerance.[3b] The introduction of organogel components into the hydrogel system inhibits the ice crystallization process. Similarly, Liu et al. employed a binary water/ethylene glycol system to fabricate anti-freezing hydrogels.[4b] This solvent could interact with polyvinyl alcohol (PVA) chains to establish extensive hydrogen bonds and induce PVA crystallization. The resulting hydorgels still keep flexible and strain-sensitive at -55 °C. Very recently, Suo group prepared polyacrylamide-alginate double network hydrogel and then equilibrated the hydrogel with $CaCl_2$ solution to depress the freezing point.[6b] At -57 °C, the dual physically cross-linked hydrogels still have fracture toughness of 5000 J/m$^2$. A stretchable ionic touch sensor was made and it can work over a large temperature range. Zhou et al. fabricated Ca-alginate/polyacrylamide hydrogels that remain unfrozen and mechanically stable even at -70 °C.[4a] The development of freeze-tolerant hydrogels is a hotspot in material science fields.

Cellulose is promising substitute to petroleum-derived synthetic polymer because of its biocompatibility, non-toxicity property and web-like entangled structure.[7] The abundant hydroxyl groups in cellulose facilitate the hydrogel formation process and interests in the fabrication of cellulose hydrogels have dramatically increased. Cellulose hydrogels can be easily prepared through a cellulose solution via physical cross-linking.[8] However, cellulose is difficult to be dissolved in common solvents due to the stiff molecular chains and high extended hydrogen bonded structure. The major challenge for preparing cellulose hydrogel is a lack of appropriate dissolution system.[9] Recently, some systems such as N-methylmorpholine-N-oxide, ionic liquids, and alkali/urea providing chances for the preparation of cellulose hydrogels.[10] So far, there is no relevant reports on anti-freezing cellulose derived hydrogels to the best of our knowledge. In prior studies, our research group reported an effective cellulose dissolution method based on inorganic salt system (zinc chloride/calcium chloride).[11] Utilization of inorganic salt solutions to solubilize and cross-link cellulose is cost-effective, recyclable and certainly stands out all among the other available solvent systems. Benefiting from the inorganic salt system of our group, we fabricated flexible and transparent cellulose membranes for gas separation application. Interestingly, this designed dissolution strategy provides a facile way to immobilize zinc ions into cellulose membrane matrix, which promote the gas separation performance.

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Combining the colligative property of ionic compounds and green zinc chloride/calcium chloride dissolution system, it is anticipated that cellulose hydrogels prepared with this method have specific functionality. The presence of inorganic salts can inhibit the ice crystallization of the hydrogel component. Herein, cellulose hydrogels that features freezing tolerance was fabricated from this binary $ZnCl_2$/$CaCl_2$ solvent system. To improve the coagulation rate of cellulose, glycerol or water was added. Besides, the noncovalent hydrogen bond cross-links and reversible metal ion-cellulose coordination endow the hydrogels with intriguing ion conductivity, remarkable remoldability and good mechanical performance.

*Construction of gels*. In this study, hydrogels were prepared from cellulose dissolved in $ZnCl_2$/$CaCl_2$ solvent system. Cotton cellulose was used as the starting material. We first prepared a solution by dissolving inorganic salts in deionized water at 75 $^oC$. Then cotton linter pulp was added and stirred vigorously, followed by sonication to remove the air bubbles. After entirely dissolved, the homogeneous solution was poured into tailor-made moulds at room temperature to prepare gels (denoted as Gel-0). (The detailed synthesis procedures are described in Experimental Section) Generally, the strong self-aggregation force caused by the extensive hydrogen bonds makes the cellulose dissolution a very difficult process. **Figure 1a** depicts the dissolution process and mechanism of cellulose in inorganic salt system. Particularly, the $ZnCl_2$ hydrate exists in the form of ionic liquid ($[Zn(OH_2)_6][ZnCl_4]$), which has extraoridinary hydrogen bond donating ability. The zinc species would compete with the protons for the hydroxyl groups in the cellulose chains. The intrinsic hydrogen bonded networks are weakened and the cellulose chains become flexible. Water molecules penetrated into the cellulose sheets and led to cellulose solubility.[12] Subsequent addition of calcium ions into the system would connect the adjacent Zn-cellulose junction zones through cooperative interactions.[13] $Zn^{2+}$ ions contribute to the cellulose dissolution and $Ca^{2+}$ ions act as gelling agent. This ionic cross-linking strategy has been applied to obtain $Ca^{2+}$/alginate and $Fe^{3+}$/cellulose hydrogels in prior studies.[14] However, the gelation of cellulose proceeded quite slowly and it took more than 48 h for the complete gelation (Gel-0). This is due to the gelling agent of $Ca^{2+}$ ions cannot effecitively lock the non-rigid Zn-cellulose chains.

Two different reagents (water and glycerol) were employed to promote the coagulation of cellulose. Certain amounts of deionized water was dropped into the cellulose/salt solution and the geletion rate was substantially enhanced. The transformation from viscous sol to hydrogel (Gel-water) with strength took place less than 10 seconds. Time dependence of the aggregation behavior during hydrogel formation can be well explained by the competition between the phase separation and the association of the molecular chains.[15] When the cross-link process (between water and cellulose chains) precedes the phase separation process, hydrogels are obtained. In the case of water-coagulated Gel-water, the phase separation is inhibited by the addition of water. By calculating the molar ratios of zinc ions and cellulose, it is found that the extra water promote the coordination equilibrium of zinc ions. As illustrated in **Figure 1a**, the hydroxyl groups in cellulose chains are coordinated to $Zn^{2+}$ ions. Five additional water molecules filled the first hydration shell. The ring hydroxyls shaded blue region would participate as a second hydration shell. It suggests that if the quantities of water fully fill the solvent's second hydration shell, gel structure is constructed. Thus, the extra water molecules lock the Zn-cellulose chains and facilitate the entanglement of cellulose and gel formation. In other words, the cross-linking dentisy of hydroxyl groups is a vital factor that affected the gel formation. To verify the proposed mechanism, we explored glycerol as gelling agent because of its excellent hydrogen bond donating ability. When adding glycerol in the dissolution process, the gelation occurred quickly in less than 30 seconds (Gel-glycerol). More importantly, glycerol may improve the anti-freezing performance as glycerol can be used as cryoprotectants to protect the protein crystals from icing at temperatures below 0 $^oC$. **Figure 1b** displays the components and fabrication process of three cellulose hydrogels.

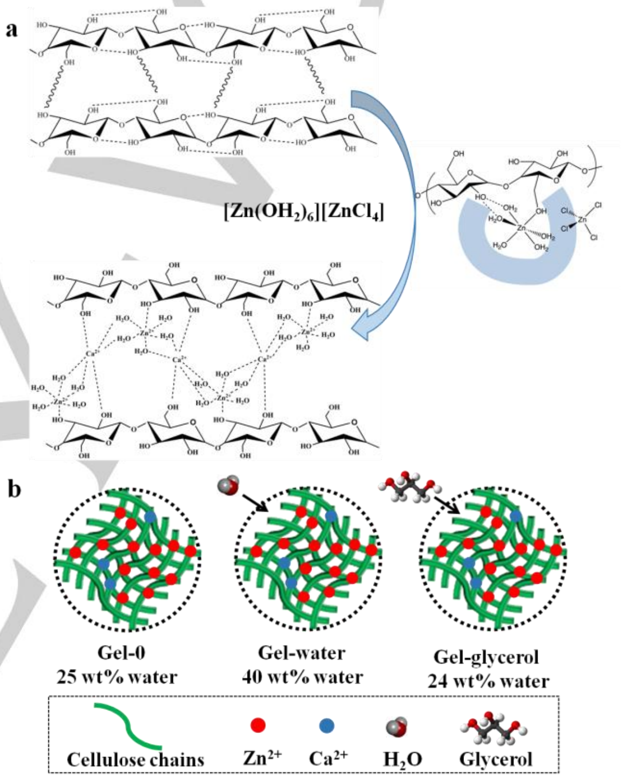


**Figure 1**. Schematic illustration of (a) the dissolution process of cellulose in $ZnCl_2$/$CaCl_2$ system and (b) the preparation and components of three cellulose hydrogels (Gel-0, Gel-water, Gel-glycerol).

*Characterization of gels.* Appearance of the cellulose hydrogels is shown in **Figure 2**. **Figure 2a** illustrates the thermal reversibility of the hydrogels. Thermal reversible sol-gel transition can be observed when gels suffered heating to cooling cycle, and it allows the gels to easily change their shapes by injecting molten cellulose into molds (**Figure 2b**). The hydrogels would turn into sol state when being heated over 60 $^oC$. After cooling to room temperature, the hydrogels changed into solid state and the new shape was realized. The remoldability is derived from the non-covalent cross-links (mainly the hydrogen-bonding interactions and the metal-cellulose interactions).[16] The temperature sensitive properties of the cellulose hydrogels also inspired us to investigate their three dimensional (3D) printing processability. As expected, 3D printing can be easily realized (Figure S1, supporting information).

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All hydrogels are transparent and colorless under visible light with a water content that ranged from 24% to 40%. The transparency proves high dispersibility of cellulose nanofibrils and metal ions in the hydrogel matrix. The cellulose nanofibrils in gel are smaller than the light wavelength, thus restricting light scattering. The light transmittance spectrum was recorded (Figure S2, supporting information) and the three distinct hydrogels follow the order of Gel-water (84%), Gel-0 (75 %) and Gel-glycerol (54%). The lower transmittance of glycerol-coagulated sample demonstrates the cellulose molecular chain packed more densely. Generally, hydrogels will lose their elasticity at temperature below 0 $^{o}$C due to the water has frozen into ice crystals. **Figure 2c** displays the good flexibility of gels at -10 $^{o}$C. It is able to withstand large knotted stretching deformation and shows good shape-recoverable ability (Video S1, supporting information). The hydrogels also have a good resistance to drying due to the coordination equilibrium between the ions and the hydroxyl groups over cellulose chains.

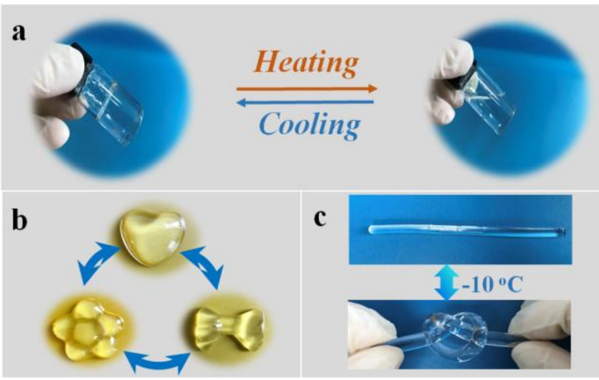


**Figure 2.** Appearance of the cellulose hydrogels. (a) Photographs indicating the thermal reversibility of Gel-glycerol. (b) Photographs showing the remoldability of Gel-glycerol. (c) Long shape Gel-glycerol demonstrating the good stretchability at -10 $^{o}$C.

The tensile properties of cellulose gels were further explored at different temperatures. As illustrated in **Figure 3a**, at room temperature (25 $^{o}$C), the tensile strength, tensile elastic modulus and elongation at break of Gel-0 are 0.28 MPa, 0.50 MPa and 120%, respectively. At the same temperature, Gel-water possesses a comparable elastic modulus of 0.49 MPa, a higher tensile strength of 0.33 MPa and a lower elongation at break of 100% compared with Gel-0. The tensile strength enhancement with the addition of water is ascribed to the higher cross-linking density.[14] The addition of extra water increases the number of hydroxyl groups and thus enhances the interaction between inorganic ions and cellulose network. The lower elongation at break of Gel-water than that of Gel-0 reveals that the introduction of water coordinate with the Zn ions and lock the hydrogel network. Therefore, the rigidity of network increases and the elasticity decreases. Physically cross-linked hydrogels are mechanically weak and soft due to the lack of strong interaction. The Gel-0 and Gel-water fabricated with cellulose are mechanically adequate for ueses as soft tissue replacements due to the high density of cross-links.

However, Gel-glycerol had poor mechanical performance (the tensile strength of 0.14 MPa, the tensile elastic modulus of 0.10 MPa and 43% elongation at break). The glycerol molecules are larger than water molecules, they can just interact with cellulose chains and have no influence on zinc ions. When temperature decreased to -20 ºC, the mechanical properties are comparable as those at 25 ºC, indicating all gels have excellent anti-freezing property. Further decreasing the temperature to -60 ºC, the thensile strength of three gels increased slightly and elongation at break decreased, indicating the ice crystals are beginning to form inside the hydrogel network. The cellulose hydrogels can be cooled to temperatures as low as -70 °C without freezing. Dynamic Scanning Calorimetry (DSC) experiments were performed to reveal the boundaries of the phase regions (Figure S3, supporting information). The aqueous $ZnCl_2$ solution has a ice point of -73 °C, while the ice point of the three gels shifted to a lower temperature. This shift is ascribed to the interactions of $ZnCl_2$ with the cellulose chains.

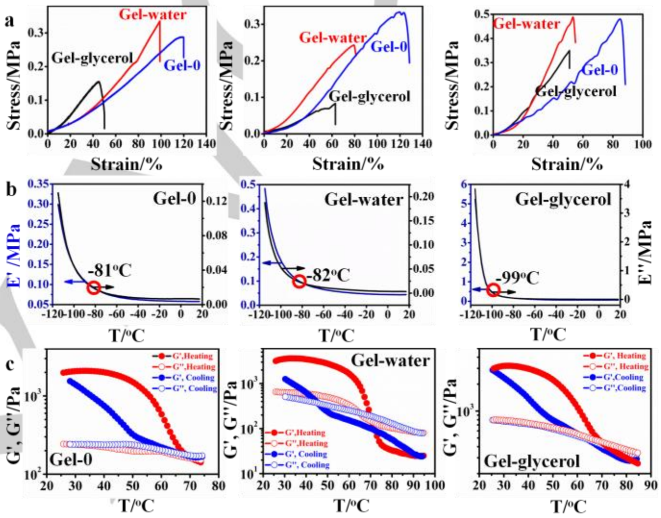


**Figure 3.** Mechanical properties of the cellulose hydrogels at different temperatures. (a) Tensile stress-strain curves of the cellulose hydrogels at 25, -20 and -60 °C. (b) Dynamic mechanical analysis (DMA) as a function of temperature. (c) Temperature dependence of the storage modulus G′ and loss modulus G″ measured in heating-to-cooling thermal cycle.

**Figure 3b** shows dynamic mechanical analysis (DMA) results of three gels. The measurement was conducted at a temperature range of 20 to -120 °C. The values of the storage modulus (E′) and loss modulus (E″) of three gels were found to maintain steady until the temperature is decreased to certain degree. For Gel-0, the E′ and E″ values increased abruptly when the temperature was decreased into -60 °C. Gel-water has the same tendency as the storage and loss moduli changed significantly around -60 °C. These results are consistent with low temperature mechanical test results, at which ice crystals start to appear around -60 ºC. By contrast, the glycerol-containing hydrogel can reduce the turning point from -60 °C to -100 °C, revealing that glycerol can improve the freezing tolerance of cellulose gels. Generally, glycerol is a non-ionic kosmotrope that constructs hydrogen bonds with water molecules, competing with water-water hydrogen bonds.[17] This interaction disrupts the formation of ice crystals in the hydrogel network.

To further elucidate the thermal reversible sol-gel transition, dynamic shear moduli were measured from 25 to 85 °C in a heating-to-cooling cycle. As displayed in **Figure 3c**, the loss modulus (G″) value is higher than storage moludus (G′) at higher temperatures for all gels. All gels exist in sol state above 70 $^{o}$C. At lower temperatures, G′ of three gels is higher than their G″, which suggests gel state. The gel-sol transition facilitated by temperature regulation. There is a crossover of G′ and G″ for three gels in heating and cooling cycle. This is a sol-gel transition temperature which is defined as $T_{gel}$. For the heating process, the $T_{gel}$ is 69, 71 and 70 °C for Gel-0, Gel-water and Gel-glycerol, respectively.[18] The physical cross-links of hydrogen bonding and crystalline domains are broken and re-built during the thermal cycle. Re-establishment of equilibrium is achieved at the cooling process. The Gel-0 has $T_{gel}$ of 68 °C, while $T_{gel}$ of Gel-water and Gel-glycerol is 45 and 65 °C, respectively in cooling cycle. The above results demonstrate thermo-induced transformation of three gels, which endows the hydrogels with remoldability and 3D-printing ability. The gel structure formation is induced by the strong self-aggregation force of cellulose and temperature can influence the balance of hydrogel.

Three gels were subjected to four successive compression loading-unloading cycles at the strain of 50% at 25 °C and -60 ºC to investigate their fatigue resistance (**Figure 4a** and **Figure 4b**). In four successive cyclic small strain deformation compression tests, the Gel-0 exhibited a large hysteresis loop with dissipated energy of 55 kJ/$m^3$ at the 1st loading-unloading cycle (25 ºC). The hysteresis loops decreased slightly after the 1st loading cycle but retained constant in the following successive three compression cycles. When the temperature decreased to -60 ºC (**Figure 4b**), the hysteresis loops for all gels in four successive cycles showed the same tendency, suggesting all gels exhibit excellent fatigue resistance. These results demonstrate that the physical cross-linking in cellulose gel network could be temporarily destroyed and rapidly recovered with small strain deformation by adjusting temperature. Besides, the Type-II cellulose crystallite hydrates reversible physically cross-links in the hydrogel network also contributed to the recovery behavior.

*Ion conductivity of gels.* Conductive hydrogel emerges as a intriguing class of self-healing flexible electronic device that are important for various applications. As shown in **Figure 4c**, the ionic conductivity of three hydrogels could reach to 7.49 S/m, which exceeds the conductivities of most conducting electrolytes. To further demonstrate the conductivity of gels, a battery powered circuit composed of a LED indicator with the hydrogel as the conductor was constructed (Video S2, Supporting information). The LED indicator displayed high brightness when employed a 5V driving voltage. The conductive capacity originates from the existence of high content of $Zn^{2+}$ ions. In addition, the cellulose chains can act as dispersion medium for metal ions, which promoted the good conductivity. Once the cellulose hydrogel is divided into two pieces, the light is extinguished. Then the two isolated parts can be re-contacted when heated to 60 °C. After cooling to room temperature, the hydrogel healed completely with a smooth and indistinguishable interface in less than 30 seconds. The light would be lighted up again under the circuit. The underlying self-healing mechanism is the dynamic dissociation and re-association of metal ion-cellulose coordination and non-covalent hydrogen bonds. During heating, hydrogen bonds dissociated and crystalline domains melted. When cooling again, new crystalline domains and hydrogen bonds recombined at the interface of two parts. The temperature-triggered healing ability promise the flexible and mechanically stable cellulose hydrogels great potential. The expected working temperature of these hydrogels ranges from -70 to 50 °C.

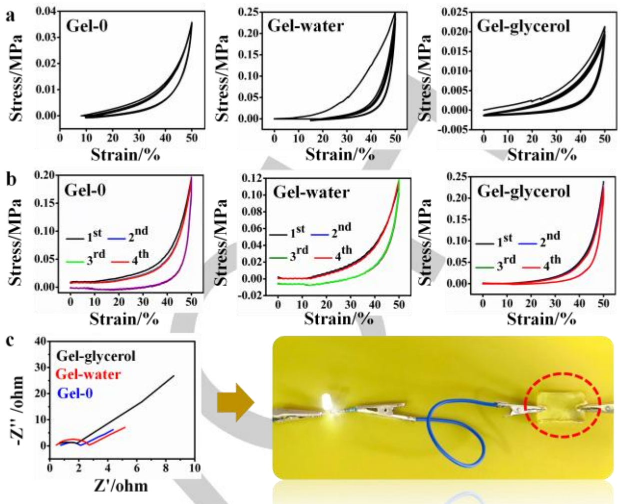


**Figure 4.** (a) Compressive strain curves at 25 °C. (b) Compressive strain curves at -60 °C. (c) The conductive propertity of the hydrogels.

The development of a facile process to prepare cellulose hydrogels with the ability to operate under harsh environmental conditions remains challenging. In this work, we report our design and synthesis of hydrogels that are composed of nanoscale cellulose nanofibers by dissolving cotton liner pulp in inorganic salts solution. The resulting cellulose hydrogels that retain their stretchability, toughness, flexibility and conductivity at temperature as low as -70 °C. Besides, the intelligent hydrogels are sensitive to the temperature, endowing the anti-freezing hydrogels with self-healing ability, remoldability and reusability.

## Experimental Section

*Materials.* Cellulose (cotton linter pulp, 1.2 wt%) with α-cellulose content beyond 96% was provided by Nantong Cellulose Fibers Company.. Zinc chloride ($ZnCl_2$, 99.0%), calcium chloride ($CaCl_2$, 99.0%) and glycerol (99.9%) were purchased from Sinopharm Chemical Reagent Company. All of the reagents were used as received unless otherwise noted.

*Hydrogels Preparation.* A two-step strategy was applied to prepared cellulose hydrogels. Initially, 9.87 g of $ZnCl_2$ and 0.2 g of $CaCl_2$ were dissolved in 3.63 g of deionized water at 75 °C to form a uniform solution. Then 0.2 g of cotton linter pulp was added and stirred vigorously for 30 min, followed by sonication to remove the bubbles. The mixture was poured into tailor-made moulds at room temperature to achieve the gels. The as-prepared hydrogel was denoted as Gel-0. As a reference, the hydrogel prepared with extra water was marked as Gel-water. 3 g of deionized water was added in the second step before the sonication treatment. The hydrogel containing glycerol was synthesized with the same procedures and designated as Gel-glycerol. 0.737 g glycerol was added in the first step after the formation of uniform solution. The proper amount of water and glycerol addition makes the gels have excellent physical and chemical properties.

*Mechanical Tests.* A minimum of three specimens were used for all test conditions. The tensile tests of gel specimens (11 and 4 mm in width and thickness) were conducted by electromechanical universal testing machine (CMT850) with a stretching rate of 10 mm/min. The compressive mechanical properties were evaluated by using the same testing system at a crosshead speed of 20 mm/min. The gels were prepared into dumbbell shaped specimens (22 and 15 mm in diamater and height, respectively) for compression and tensile testing. The elastic modulus was calculated in the range of 5-10% strain from the loading curve.

*Conductivity Measurements.* Ionic conductivity of the gels was measured in a Swagelok cell with two parallel stainless steel electrodes by the electrochemical impedance spectroscopy (EIS) method. Impedance data were obtained under an electrchemical working station CHI660E (Chenhua, Shanghai) in the frequency range of 0.01 Hz to 100 kHz at room temperature. The ionic conductivity of hydrogels can be calculated from Eq. (1) from the known area and thickness of the hydrogels membranes:

$$\sigma_H = \frac{L}{R_b A} \quad (1)$$

Where L is the gel membrane thickness (cm), A is the activity area ($cm^2$) and $R_b$ is the resistance of the hydrogel membrane.

*Dynamic Mechanical Analysis (DMA) measurement.* Elastic and loss modulus of the gels were measured using a DMA 242 C/1/G (Netzsch, Germany) in compression mode (10 Hz) over a temperature range of 20 °C to -120 °C with a cooling rate of 3 °C/min, a load of 2 N, and an amplitude of 20 m. Measurements were taken at a strain γ = 1% and frequency ω = 6.28 rad/s with heating and cooling rates of ±0.5 °C/min.

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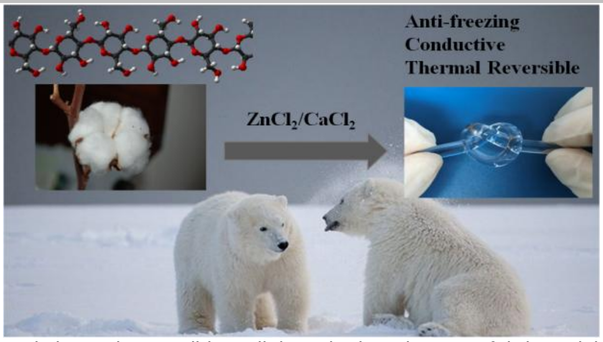
## Entry for the Table of Contents

# COMMUNICATION




Anti-freezing and thermal reversible cellulose hydrogels were fabricated by a specially designed inorganic salt system. This new type of cellulose based hydrogels is anticipated to construct flexible and wearable devices for services at temperature as low as -70 °C.

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